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Long-term functionality of woody debris structures for forest-floor small mammals on clearcuts

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ABSTRACT

Restoration practices are much needed on clearcut openings in commercial forest landscapes where some mammal species have declined in abundance from a loss of preferred food, cover, and other components of stand structure. Retention of excess woody debris in piles and windrows provides habitat for forest-floor small mammals and some of their predators such as small mustelids. However, it is unknown if these retention habitats are used over longer periods (> 10 years) as new forests grow and develop on harvested sites, or do they become unoccupied? We tested the hypotheses (H) that (H_1) abundance, species richness, and diversity of the forest-floor small mammal community, and (H_2) reproduction and survival of the southern red-backed vole (*Myodes gapperi*), long-tailed vole (*Microtus longicaudus*), and deer mouse (*Peromyscus maniculatus*) would decline on sites with woody debris structures, compared with sites of dispersed woody debris or uncut forest, up to 12 years post-construction in south-central British Columbia, Canada.

Woody debris structures provided habitat on new clearcuts for *M. gapperi* at comparable or higher abundance than in uncut forest and 5.0 to 7.6 times higher than on dispersed sites in the first 5-year period. Although numbers declined after the initial three years, populations of *M. gapperi* in debris structures recovered to earlier abundance levels at 11–12 years post-construction. Mean abundance of *M. longicaudus* was consistently higher (2.8 to 3.5 times) in piles and windrows than in sites with dispersed woody debris over the first 5-year period. Populations of *M. longicaudus* were high in all three treatment sites at 11 years post-construction reaching mean annual peak numbers of 24, 42, and 36 voles per ha in the dispersed, piles, and windrow sites, respectively. Mean abundance of *P. maniculatus* was similar among treatment sites and consistent over time. Mean abundance of total small mammals was consistently higher (1.8 to 2.4 times) in piles and windrows than dispersed or forest sites in the first 5-year period and this pattern was continued at 11–12 years post-construction. At 11 years post-construction, all treatment sites had the highest peak numbers per ha in the study: dispersed (40.3), piles (64.1), windrows (56.1), and forest (29.0). Our results did not support H_1 as abundance, species richness, and diversity were increased or maintained in the debris structures over the 12-year period. Reproduction and survival followed the pattern of abundance for the major species, and hence H_2 was not supported. Our study is the first to measure long-term (up to 12 years) responses of forest-floor small mammals to constructed piles and windrows of woody debris as a means of habitat retention on clearcuts. These mammalian species, particularly voles, may then serve as prey for marten and other mustelids. This relationship provides further support for piles and windrows to act as baseline trophic structures in ecological restoration of cutover forest land.

1. Introduction

Maintenance of mammal diversity, as a component of biodiversity, is a major conservation goal in commercial forest landscapes. Coarse woody debris (dead or down wood) on the floor of coniferous and deciduous forests contributes to biodiversity through structures and functions that are essential for long-term ecosystem productivity such as nutrient cycling, substrates for other plants and fungi, and modification of microclimate (Harmon et al., 1986; McComb and Lindenmayer, 1999; Bunnell and Houde, 2010). In addition, woody debris provides crucial habitat for a wide variety of forest mammal species in terms of foraging, resting, reproduction, and thermal and security cover (Maser et al., 1979; Carey and Johnson, 1995; McComb, 2003).

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Forest harvesting often leaves excess woody debris on the forest floor and retention of some debris may provide critical habitat for wildlife (Fauteux et al., 2012; Sullivan et al., 2012). Woody debris structures may serve as a building block for trophic structure in ecological restoration of forests (Montoya et al., 2013; Fraser et al., 2015). This retention practice may be particularly relevant on large clearcut openings (e.g., > 10 ha) in conventional, but also larger (e.g., > 100 ha) salvage harvesting operations (Lindenmayer et al., 2008), where some mammal species have declined in abundance from a loss of preferred food, cover, and other components of forest stand structure (Fisher and Wilkinson, 2005).

Furbearers, American marten (*Martes americana*) in particular, are negatively affected by clearcutting and practices to retain habitat attributes that last for several decades are urgently needed (Thompson, 1994; Hargis et al., 1999; Lavoie et al., 2019; and others). Small mustelids (marten; short-tailed weasel, *Mustela erminea*; long-tailed weasel, *M. frenata*) avoid large openings as they may be prey species for other carnivores (Buskirk and Zielinski, 2003). Similarly, the old forest-dependent species of small mammals such as the red-backed vole (*Myodes gapperi*), a major prey species of mustelids (Martin, 1994), also disappear soon after clearcutting and it is unclear when they will return to pre-harvest levels of abundance (Zwolak, 2009). In North America, several generalist forest-floor species such as the deer mouse (*Peromyscus maniculatus*), northwestern chipmunk (*Neotamias amoenus*), *Microtus* voles, and *Sorex* shrews occupy a variety of habitats and often persist on clearcuts for variable periods (Fisher and Wilkinson, 2005; Zwolak, 2009). Voles (*Myodes* and *Microtus* spp.), in particular, are major prey species for several mammalian carnivores such as marten (Martin, 1994), short-tailed weasels, and long-tailed weasels (Simms, 1979; Buskirk and Zielinski, 2003; Wilk and Raphael, 2017). Both genera of microtines may undergo dramatic fluctuations in abundance over many years and decades (Boonstra and Krebs, 2012; Krebs, 2013).

A windrow or series of piles constructed from woody debris may connect patches of mature forest and riparian areas to allow small mammals and some of their predators (e.g., small mustelids) to use clearcut openings (Seip et al., 2018; Sullivan et al., 2012, 2017a). Responses in species richness and diversity of forest-floor small mammals also followed this pattern. However, relative abundance levels of voles tended to decline by the third year in studies, to date, but these declines may have been related to multi-annual changes in vole abundance patterns. No data exist to determine if these structures are used over longer periods as new forests grow and develop on harvested sites. Indeed, studies on long-term (+10 years) responses of wildlife to constructed habitats after forest clearcutting are essentially non-existent. Not surprisingly, management of woody debris in forest ecosystems would seem crucial to conservation of biodiversity, but suffers from a dearth of experimental studies in both Europe and North America (Seibold et al., 2015).

Woody debris tends to dry out with the settling of fine-scale needles and twigs and consequent shifting of larger debris during the first 5–10 years post-harvest on clearcuts. The first phase of decomposition is often slow as the wood becomes colonized and develops a moisture status suitable for decomposer organisms (McComb, 2003; Laiho and Prescott, 2004). Woody debris within forests presumably has ongoing decay processes to enhance food sources such as fungi, arthropods, and microarthropods for small mammals (Maser et al., 1979; Maser and Trappe, 1984; Seastedt et al., 1989; Amaranthus et al., 1994; Jacobs and Work, 2012). Depending on the ecological zone, degree of decay of woody debris depends on temperature and moisture regimes, microarthropod abundance, size of debris, and successional stage of the regenerating forest (Laiho and Prescott, 2004).

Thus, we ask if woody debris structures, originally created on clearcuts, become abandoned habitats for small mammals, and consequently, their predators, or will these retention habitats remain viable through time in the regenerating forest? We investigated the long-term (up to 12 years post-construction) responses of small mammals in piles and windrows of woody debris in clearcut areas. We tested the hypotheses (H) that (H_1) abundance, species richness, and species diversity of the forest-floor small mammal community, and (H_2) reproduction and survival of the major species *M. gapperi*, *M. longicaudus*, and *P. maniculatus*, would decline over time in piles and windrows of woody debris, compared with sites of dispersed woody debris or uncut forest.

2. Methods

2.1. Study area

The study area was located in south-central British Columbia (BC), Canada, in the Bald Range 25 km west of Summerland (49⁰40′N; 119⁰53′W) in the upper Interior Douglas-fir (IDF_{dk}; d,k = dry precipitation regime, cool temperature regime) and Montane Spruce (MS_{dm}; d,m = dry precipitation regime, mild temperature regime) biogeoclimatic subzones (Meidinger and Pojar, 1991). Topography was rolling hills at 1400 to 1520 m elevation. The upper IDF and MS have a cool, continental climate with cold winters and moderately short, warm summers. Mean annual precipitation ranges from 30 to 90 cm. Open to closed mature forests of Douglas-fir (*Pseudotsuga menziesii*) cover much of the IDF zone, with even-aged post-fire lodgepole pine (*Pinus contorta* var. *latifolia*) stands at higher elevations. Hybrid interior spruce (*Picea glauca* × *P. engelmannii*) and subalpine fir (*Abies lasiocarpa*) are the dominant shade-tolerant climax trees. Trembling aspen (*Populus tremuloides*) is a common seral species and black cottonwood (*Populus trichocarpa*) occurs on some moist sites (Meidinger and Pojar, 1991).

Prior to harvesting, study stands were composed of a mixture of lodgepole pine with variable amounts of Douglas-fir, interior spruce, and subalpine fir. Average ages of post-fire lodgepole pine ranged from 80 to 120 years and for Douglas-fir and other conifers ranged from 120 to 220 years. Average tree heights ranged from 10.5 to 19.5 m for lodgepole pine and from 16.7 to 27.5 m for Douglas-fir and other conifer species.

2.2. Experimental design

The study was a randomized complete block design (based on topography and geographic location) with 3 replicates each of woody debris distributed: (i) uniformly over each site; (ii) into several "piles" with an average (± SE) of 8.8 ± 2.3 per ha; and (iii) into "windrows" with an average (± SE) of 6.3 ± 1.8 per ha; and (iv) uncut old-growth/mature forest. The 12 sites (4 treatments × 3 replicates) were selected on the basis of operational scale, harvest sites that were the size of typical forestry operations, and reasonable proximity of sites to one another within the study area. Each of the 9 woody debris sites was located on a separate clearcut unit; these sites ranged in mean (± SE) size (ha) from 6.1 ± 1.0 (dispersed) to 7.6 ± 1.7 (piles) to 8.7 ± 1.3 (windrows). All sites were separated by an average (± SE) of 0.77 ± 0.29 km (range 0.2–3.0 km) to enhance biological and statistical independence. A measure of this independence was that no individual vole was captured on more than one sampling grid.

2.3. Woody debris treatments

All timber harvesting was targeted at lodgepole pine salvage after, or impending, mountain pine beetle (MPB) (*Dendroctonus ponderosae*) attack. The harvesting system was clearcut with some single-tree retention of Douglas fir and spruce. Harvesting and subsequent woody debris treatments were installed in October 2006 where debris piles and windrows were created during processing of cut timber, followed by some specific site preparation work with an excavator. Structures were composed of tops, branches, and bole ends of harvested trees, as well as trees knocked down during harvest, low-quality commercial trees, dead wood, and non-commercial trees left at the harvest site. The largest

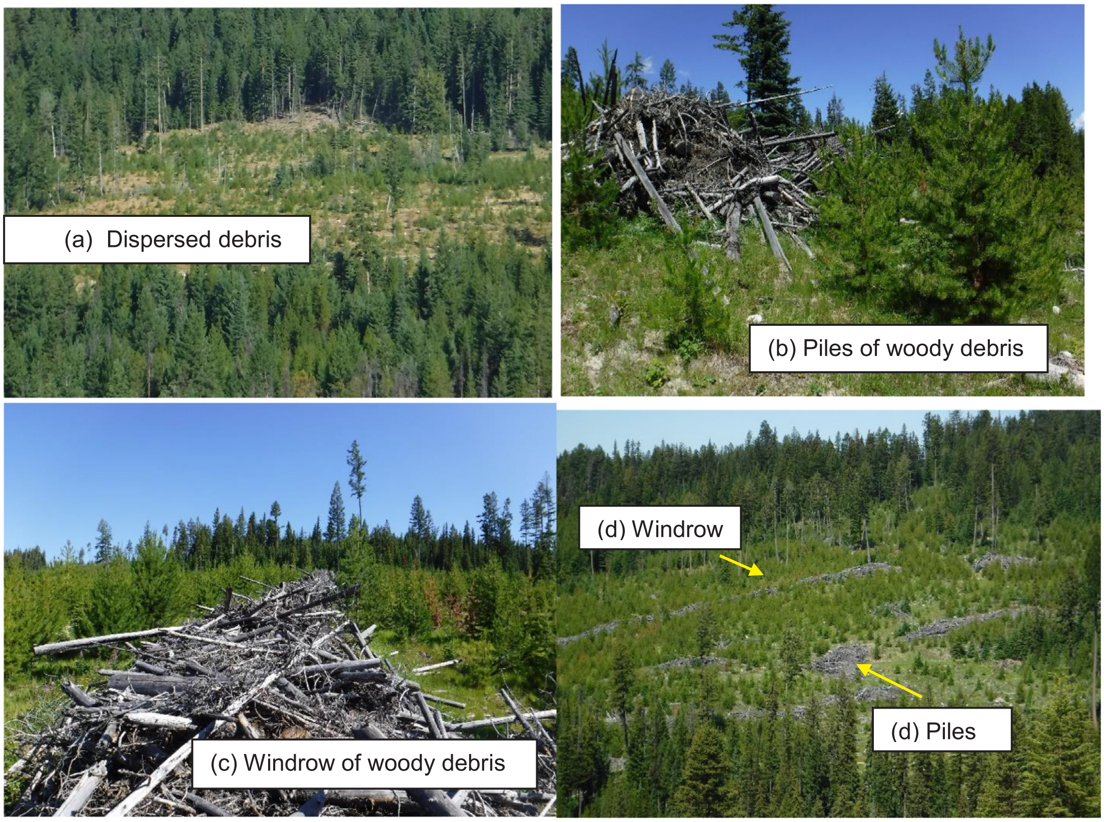

Fig. 1. Photographs of post-harvest woody debris sites at the study areas in south-central British Columbia, Canada: (a) dispersed, (b) piles, (c) windrow and (d) connectivity of piles and a windrow across a forest plantation, in August 2018, 12 years after construction.

material retained in structures averaged 20 to 30 cm in diameter. Debris piles and windrows averaged 2–3 m in height and 5–8 m in diameter or width; and were located near landings where log processing occurred. The calculated dry mass (t/ha) of woody material was 71.50 for piles and 62.15 for windrows; with a higher than typical volume of stems and thick branches left on site owing to mortality of pine trees by MPB (Sullivan et al., 2011). The three original woody debris treatments are illustrated in August 2018, 12 years after construction (Fig. 1a–d).

Location of sampling grids for forest-floor small mammals (see below) on dispersed debris sites, and those that incorporated piles or windrows, were at comparable distances (0.1 – 0.2 km, on average) to the edge of each respective clearcut site. Edge habitats were either mature/old-growth forest or 10- to 20-year-old second-growth forest.

2.4. Forest-floor small mammals

Forest-floor small mammal populations were sampled at 4-week intervals from May to October 2007, 2008, and 2009; May or June to September 2010 and 2011; and June to September or October 2017 and 2018. One live-trapping grid (1 ha) was located in each of the 12 sites and had 49 (7 × 7) trap stations at 14.3-m intervals with 1 or 2 Longworth live-traps at each station. Number of traps at a station was increased when trap occupancy was greater than 60%. Traps were supplied with whole oats and carrot, with cotton as bedding. Each trap had a 30-cm × 30-cm plywood cover for protection from sunlight (heat) and precipitation. Traps were set on the afternoon of day 1, checked on the morning and afternoon of day 2 and morning of day 3, and then locked open between trapping periods. All small mammals captured were ear-tagged with serially numbered tags, assessed for breeding condition, weighed on Pesola spring balances, and point of capture recorded. Duration of breeding season was noted by palpation of male testes and the condition of mammaries of the females (Krebs, 2013). A pregnancy was considered successful if a pregnant female captured in one month was recaptured lactating in the following month. Animals were released on the grids immediately after processing. Unfortunately, there was a high mortality rate for shrews in the traps overnight, but this was unavoidable in practice. Therefore, shrews were collected, frozen, and later identified according to Nagorsen (1996). All handling of animals was in accordance with the principles of the Animal Care Committee, University of British Columbia.

2.5. Demographic analysis

Abundance estimates of the red-backed vole, long-tailed vole *(Microtus longicaudus*), deer mouse, and northwestern chipmunk were derived from the Jolly-Seber (J-S) stochastic model for open populations with small sample size corrections (Seber, 1982; Krebs, 1999). Minimum number alive was used to estimate populations of the meadow vole (*M. pennsylvanicus*) and heather vole (*Phenacomys intermedius*); number of individuals was used for the montane shrew (*S. monticolus*) and common shrew (*S. cinereus*). We calculated the effective trapped area (ETA) for each grid for each of the major species, based on mean maximum distance moved (MMDM) as a boundary strip method (Krebs et al., 2011). Estimates of population size were converted to a density estimate by dividing population estimates for each trapping period by the ETA. We consider this estimate a “density index” (Krebs et al., 2011). Jolly trappability for the major species was calculated according to Krebs and Boonstra (1984). Species richness was the total number of species sampled for the mammal communities in each site (Krebs, 1999). Species diversity was based on the Shannon-Wiener index which is well represented in the ecological literature (Burton et al., 1992). Mean annual measurements of abundance, species richness, and species diversity of small mammals were calculated using the estimated parameter for each species or community for a given sampling period and then averaged over the number of sampling periods for each year.

Table 1
Mean (n = 3 replicate sites) ± SE annual abundance for each species, total abundance, and species richness, and diversity per ha within the forest-floor small mammal community for the 2007–2011 period among the four treatments, and results of RM-ANOVA. F-values identified by * were calculated using an H-F correction factor, which decreased the stated degrees of freedom due to correlation among repeated measures. Within a row, columns of mean values with different letters are significantly different by Duncan's multiple range test (DMRT), adjusted for multiple contrasts**.** Significant values are given in bold text.

Parameter and year	Treatment				RM-ANOVA Treatment		Time		Treatment × time	
	Dispersed	Piles	Windrows	Forest	$F_{3,6}$	P	$F_{4,32}$	P	$F_{12,32}$	P
Mean abundance										
M. gapperi	B	A	A	A	8.45	**0.01**	3.61	**0.02**	1.35	0.24
2007	1.37 ± 0.58	7.07 ± 2.58	8.11 ± 1.99	6.93 ± 2.13						
2008	0.87 ± 0.35	7.69 ± 3.02	13.38 ± 1.95	5.53 ± 0.83						
2009	1.24 ± 0.86	3.83 ± 1.05	7.82 ± 2.25	2.95 ± 0.81						
2010	0.77 ± 0.44	2.10 ± 0.50	2.58 ± 1.20	3.88 ± 2.09						
2011	0.35 ± 0.24	2.48 ± 1.47	2.97 ± 1.72	7.07 ± 1.40						
M. longicaudus	B	A	A	B	10.90	**< 0.01**	3.75	**0.01**	0.88	0.58
2007	1.39 ± 0.59	10.75 ± 3.63	7.26 ± 1.73	0.00 ± 0.00						
2008	0.89 ± 0.28	8.10 ± 1.33	5.17 ± 1.91	0.00 ± 0.00						
2009	0.29 ± 0.19	4.62 ± 1.45	4.56 ± 1.88	0.00 ± 0.00						
2010	2.04 ± 1.02	5.12 ± 1.39	6.90 ± 3.81	0.00 ± 0.00						
2011	7.04 ± 2.27	12.38 ± 2.66	8.80 ± 4.34	0.00 ± 0.00						
P maniculatus					0.58	0.65	0.43*	0.73	0.33*	0.97
2007	5.92 ± 1.08	3.48 ± 0.57	3.99 ± 1.90	6.61 ± 1.95						
2008	7.48 ± 2.64	4.40 ± 0.63	2.95 ± 0.94	3.96 ± 1.17						
2009	5.80 ± 1.86	5.22 ± 0.96	3.77 ± 0.44	3.59 ± 0.53						
2010	3.99 ± 0.59	4.09 ± 0.96	3.52 ± 0.97	2.30 ± 0.26						
2011	3.54 ± 1.28	2.86 ± 1.25	5.04 ± 3.61	2.19 ± 1.53						
N. amoenus	B	A	A	B	22.57	**< 0.01**	11.52*	**< 0.01**	2.08*	0.07
2007	2.07 ± 0.32	5.40 ± 0.53	3.31 ± 0.02	1.28 ± 0.45						
2008	3.04 ± 0.90	9.47 ± 0.37	6.45 ± 0.61	1.03 ± 0.09						
2009	4.65 ± 1.24	10.23 ± 1.15	8.64 ± 0.88	1.81 ± 0.50						
2010	3.36 ± 1.19	9.32 ± 1.53	8.80 ± 1.07	1.21 ± 0.35						
2011	2.12 ± 1.00	7.18 ± 0.96	6.73 ± 0.83	1.26 ± 0.57						
Total small mammals	B	A	A	B	11.21	**< 0.01**	1.54	0.21	0.35	0.97
2007	12.92 ± 0.96	26.93 ± 1.11	23.17 ± 1.68	14.87 ± 3.17						
2008	15.79 ± 2.15	30.65 ± 1.80	29.12 ± 3.35	10.57 ± 1.19						
2009	13.61 ± 1.36	24.39 ± 2.31	26.06 ± 1.95	8.41 ± 1.24						
2010	11.68 ± 1.46	20.83 ± 1.63	22.41 ± 4.91	7.45 ± 2.66						
2011	15.45 ± 4.45	25.22 ± 2.17	24.12 ± 7.83	10.51 ± 1.98						
Species richness	B	AB	A	C	25.77	**< 0.01**	2.67	**0.05**	0.97	0.50
2007	3.89 ± 0.11	3.94 ± 0.06	3.95 ± 0.22	2.78 ± 0.11						
2008	4.44 ± 0.24	4.55 ± 0.15	4.67 ± 0.35	2.94 ± 0.06						
2009	3.17 ± 0.51	4.22 ± 0.28	4.67 ± 0.19	2.78 ± 0.11						
2010	3.47 ± 0.33	4.00 ± 0.12	4.27 ± 0.07	2.87 ± 0.07						
2011	3.83 ± 0.17	3.92 ± 0.36	4.17 ± 0.17	2.42 ± 0.46						
Species diversity	A	A	A	B	20.04	**< 0.01**	0.63	0.64	0.65	0.79
2007	1.60 ± 0.10	1.67 ± 0.02	1.64 ± 0.04	1.22 ± 0.09						
2008	1.74 ± 0.12	1.85 ± 0.04	1.80 ± 0.13	1.22 ± 0.04						
2009	1.30 ± 0.21	1.72 ± 0.09	1.89 ± 0.08	1.26 ± 0.13						
2010	1.50 ± 0.19	1.65 ± 0.04	1.73 ± 0.10	1.32 ± 0.04						
2011	1.68 ± 0.04	1.60 ± 0.11	1.64 ± 0.17	0.98 ± 0.32						

For the assessment of reproduction and survival, mass (g) at sexual maturity was used to determine age classes of *M. gapperi*: juvenile = 1–18; adult ≥ 19; *M. longicaudus*: juvenile = 1–24; adult ≥ 25; *P. maniculatus*: juvenile = 1–20; adult ≥ 21. Juveniles were considered to be young animals recruited during the study. Measurements of recruitment (new animals that entered the population through reproduction and immigration), and number of successful pregnancies were derived from the sample of animals captured in each trapping session and then summed for each summer period. Mean summer (May-June to September-October) and winter (October-November to April-May) survival rates (28-day) for the 5-year (2007–2011) and 2-year (2017–2018) periods were estimated from the J-S model.

2.6. Statistical analysis

A repeated-measures analysis of variance (RM-ANOVA) (IBM Corp., 2017) was used to determine the effect of the four treatments and two time periods (2007–2011 and 2017–2018) on mean values for abundance of *M. gapperi*, *M. longicaudus, P. maniculatus*, *N. amoenus*, total small mammals, species richness and species diversity of the forest-floor small mammal community, as well as time and treatment × time interactions. This analysis was also conducted to detect differences in mean values for number of recruits, number of successful pregnancies, and J-S summer and winter survival for *M. gapperi*, *M. longicaudus*, and *P. maniculatus*, as well as time and treatment × time interactions. A one-way ANOVA was conducted to determine the effect of the four treatments on overall mean trappability for 2007–2011 and 2017–2018, and mean J-S summer (2017) and winter (2017–18) survival for each of the major species. The low and inconsistent sample sizes (< 2/ha) of meadow voles, heather voles, and shrews precluded statistical analysis. Homogeneity of variance was measured by the Levene statistic. Mauchly's W-test statistic was used to test for sphericity (independence of data among repeated measures) (Littel, 1989; Kuehl, 1994). For data found to be correlated among years, the Huynh-Feldt (H-F) correction was used to adjust the degrees of freedom of the within-subjects F-ratio (Huynh and Feldt, 1976). Proportional data were transformed by arcsin square root (Fowler et al., 1998). Overall mean values for the two time periods ($n = 15$, 3 replicate sites × 5 years; and $n = 6$, 3 replicate sites × 2 years) ± 95% confidence intervals (CIs) were also calculated (Zar, 1999). Duncan's multiple range test (DMRT), adjusted for multiple contrasts, was used to compare mean values based on RM-ANOVA results (Saville, 1990). In all analyses, the level of significance was at least $P = 0.05$.

3. Results

3.1. Abundance of small mammals

A total of eight species of forest-floor small mammals, composed of 5108 individuals, were captured over the two study periods in 36 trapping periods. The long-tailed vole was the most common species captured with 1484 individuals followed by the red-backed vole (1154), deer mouse (1201), northwestern chipmunk (800), montane shrew (225), meadow vole (117), heather vole (107), and masked shrew (20). Mean Jolly trappability was similar ($P \geq 0.09$) among treatment sites for all three major species in both periods, thereby meeting the homogeneity assumption related to capture probabilities. Mean (± SE) Jolly trappability for *M. gapperi* ranged from 79.4 ± 8.1% to 91.3 ± 3.0%; for *M. longicaudus* ranged from 61.6 ± 7.0% to 91.4 ± 4.4%; and for *P. maniculatus* ranged from 76.5 ± 5.9% to 94.0 ± 0.7%. There were no significant treatment × time interactions for any species or other parameters in these population and community analyses.

Mean abundance of *M. gapperi* was significantly ($F_{3,6} = 8.45$; $P = 0.01$) different among treatment sites with the piles, windrow, and forest sites at consistently higher (DMRT; $P = 0.05$) numbers (5.0 to 7.6 times on average) than the dispersed sites in 2007–2011 (Table 1; Fig. 2). Populations reached a mean annual peak of 14 to 18 red-backed voles per ha in the windrow sites in 2007–2008. Mean abundance of *M. gapperi* changed significantly ($F_{4,32} = 3.61$; $P = 0.02$) with time with higher numbers during 2007–2009 than in 2010–2011 (Fig. 2). *M. gapperi* populations increased in the forest sites in 2011, and then again in 2017–2018, being 2.6 to 6.1 times higher than in the other treatment sites, but similar ($F_{3,6} = 2.26$; $P = 0.18$) statistically (Table 2; Fig. 2). Annual population peaks were 14 to 21 red-backed voles per ha in 2017–2018 (Fig. 2).

Mean abundance of *M. longicaudus* was significantly ($F_{3,6} = 10.90$; $P < 0.01$) different among treatment sites with the piles and windrow at consistently higher (DMRT; $P = 0.05$) numbers (2.8 to 3.5 times on average) than the dispersed sites in 2007–2011 (Table 1; Fig. 3). No long-tailed voles were captured in the forest sites. Mean abundance of *M. longicaudus* changed significantly ($F_{4,32} = 3.75$; $P = 0.01$) over time with numbers relatively high in 2007, declining in 2008–2010, and high again in 2011 (Fig. 3). Populations of *M. longicaudus* were high in all three treatment sites in 2017 reaching mean annual peak numbers of 24, 42, and 36 voles per ha in the dispersed, piles, and windrow sites, respectively (Fig. 3). Mean annual abundance of *M. longicaudus* followed this same pattern in 2017–2018 with the decline in numbers over time approaching statistical significance ($F_{1,8} = 4.78$; $P = 0.06$) (Table 2; Fig. 3).

Mean abundance of *P. maniculatus* was similar ($P \geq 0.59$) among treatment sites and consistent over time in both the 2007–2011 and 2017–2018 periods (Table 2; Fig. 4). Mean numbers ranged from 2.2 to 7.2 deer mice per ha in the first period and from 0.9 to 5.0 in the second period (Table 2). Mean abundance of *N. amoenus* was significantly ($F_{3,6} = 22.57$; $P < 0.01$) different among treatment sites with the piles and windrows at consistently higher (DMRT; $P = 0.05$) numbers (2.5 to 5.7 times on average) than the dispersed and forest sites in 2007–2011 (Table 1; Fig. 5). Mean numbers ranged from 6.5 to 10.2 chipmunks per ha in the piles and windrows during 2008–2010. There was a significant ($F_{4,32} = 11.52$; $P < 0.01$) effect of time owing to an initial increase and then general decline in numbers. Mean abundance of this sciurid was similar ($P = 0.17$) in the 2017–2018 period with numbers ranging from 1.2 to 4.6 (Table 2; Fig. 5).

Mean abundance of total small mammals, including the less common species, was significantly ($F_{3,6} = 11.21$; $P < 0.01$) different among treatment sites with the piles and windrows at consistently higher (DMRT; $P = 0.05$) numbers (1.8 to 2.4 times on average) than the dispersed and forest sites in 2007–2011 (Table 1; Fig. 6). Mean annual peak numbers of small mammals per ha in 2007–2008 in the piles and windrows were 49–52 and 39–44, respectively, before settling to approximately 30 animals per ha in 2009–2011 (Fig. 6). Although not statistically significant ($P = 0.14$), a similar pattern was recorded in the 2017–2018 period, but with substantially higher mean annual peak

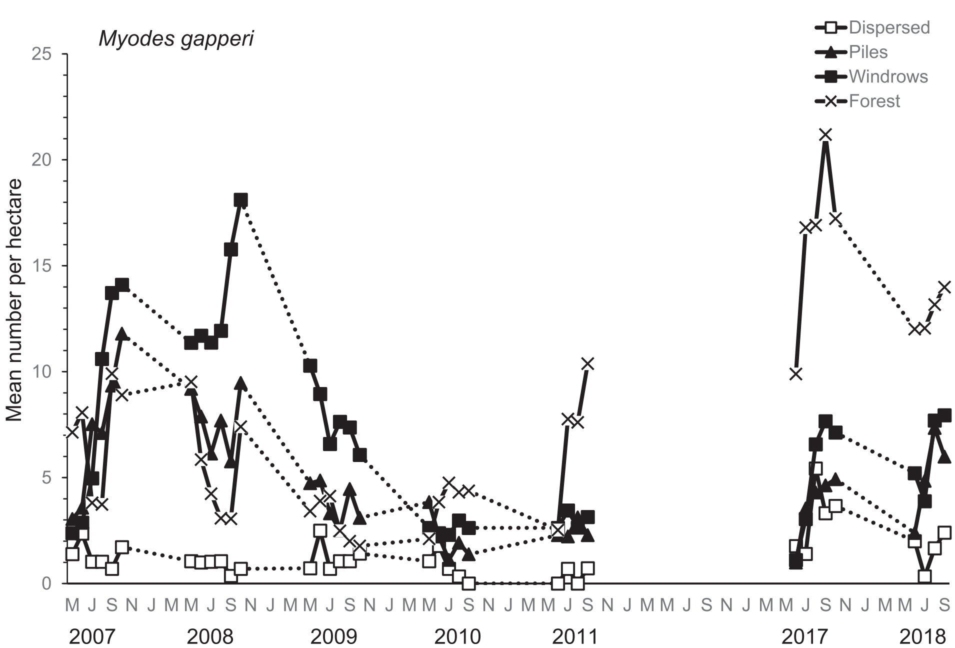

Fig. 2. Mean ($n = 3$ replicate sites) number of *Myodes gapperi* per ha as an index based on Jolly-Seber population estimates and effective trapped area in the dispersed, piles, windrow, and forest sites during 2007–2011 and 2017–2018. Data points indicate individual trapping weeks each summer (May to October) and dots indicate winter periods when populations were not sampled.

Table 2
Mean (n = 3 replicate sites) ± SE annual abundance for each species, total abundance, and species richness, and diversity per ha within the forest-floor small mammal community for the 2017–2018 period among the four treatments, and results of RM-ANOVA. Within a row, columns of mean values with different letters are significantly different by Duncan's multiple range test (DMRT), adjusted for multiple contrasts. Significant values are given in bold text.

Parameter and year					RM-ANOVA					
	Treatment				Treatment		Time		Treatment × time	
	Dispersed	Piles	Windrows	Forest	$F_{3,6}$	P	$F_{1,8}$	P	$F_{3,8}$	P
Mean abundance										
M. gapperi					2.26	0.18	0.85	0.38	2.56	0.13
2017	3.12 ± 0.48	3.68 ± 0.86	5.10 ± 1.92	16.41 ± 6.85						
2018	1.60 ± 0.57	5.14 ± 1.22	6.18 ± 1.70	12.49 ± 5.39						
M. longicaudus	**A**	**A**	**A**	**B**	5.77	**0.03**	4.78	0.06	0.63	0.61
2017	14.65 ± 5.63	23.9 ± 8.24	21.67 ± 7.93	0.00 ± 0.00						
2018	7.42 ± 1.30	11.23 ± 2.03	10.88 ± 4.99	0.00 ± 0.00						
P. maniculatus					0.70	0.59	11.44	< **0.01**	1.49	0.29
2017	2.43 ± 0.88	4.77 ± 0.96	2.84 ± 0.66	5.02 ± 1.33						
2018	2.34 ± 1.10	1.55 ± 0.07	0.85 ± 0.19	1.20 ± 0.70						
N. amoenus					2.41	0.17	2.90	0.13	1.92	0.20
2017	2.13 ± 0.86	2.40 ± 1.33	3.19 ± 0.36	2.20 ± 0.46						
2018	2.99 ± 0.20	4.18 ± 0.69	4.57 ± 0.41	1.20 ± 0.42						
Total small mammals 2017					2.66	0.14	14.44	< **0.01**	0.12	0.95
	28.45 ± 4.59	41.21 ± 7.06	37.13 ± 8.88	24.56 ± 6.38						
2018	16.59 ± 1.73	26.14 ± 1.12	24.39 ± 6.13	14.89 ± 4.75						
Species richness	**A**	**A**	**A**	**B**	13.55	< **0.01**	7.90	**0.02**	1.02	0.43
2017	5.67 ± 0.24	5.13 ± 0.41	5.53 ± 0.24	3.53 ± 0.18						
2018	4.58 ± 0.51	5.17 ± 0.44	4.67 ± 0.08	2.33 ± 0.24						
Species diversity	**A**	**A**	**A**	**B**	4.87	**0.05**	0.35	0.57	0.55	0.66
2017	1.88 ± 0.19	1.73 ± 0.30	1.84 ± 0.24	1.25 ± 0.15						
2018	1.81 ± 0.12	1.96 ± 0.16	1.71 ± 0.01	0.74 ± 0.29						

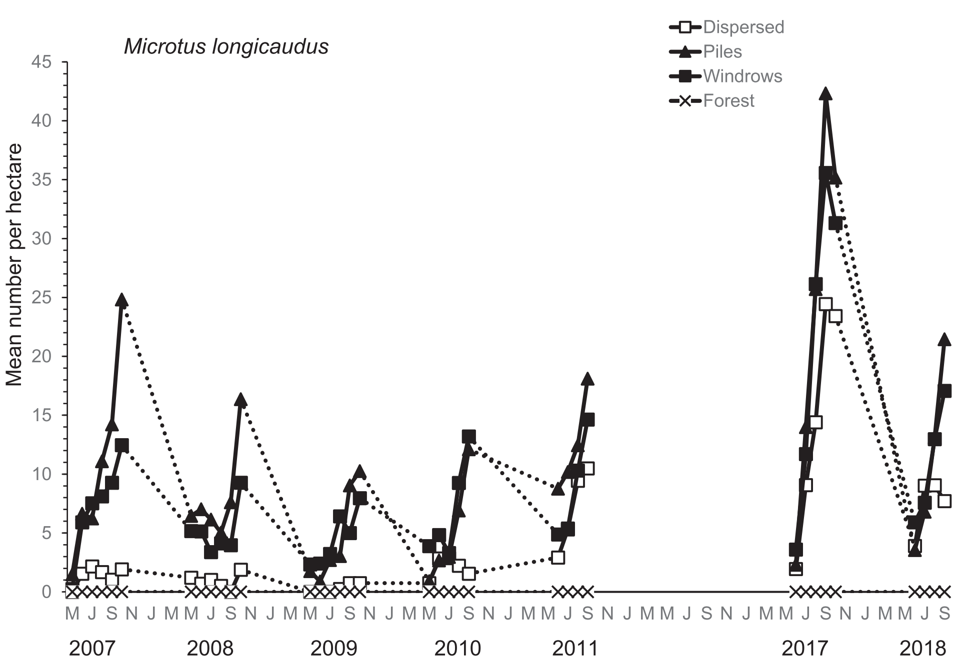

Fig. 3. Mean (n = 3 replicate sites) number of *Microtus longicaudus* per ha as an index based on Jolly-Seber population estimates and effective trapped area in the dispersed, piles, windrow, and forest sites during 2007–2011 and 2017–2018. Data points indicate individual trapping weeks each summer (May to October) and dots indicate winter periods when populations were not sampled.

numbers of small mammals (Table 2; Fig. 6). In 2017, all treatment sites had the highest peak numbers per ha in the study: dispersed (40.3), piles (64.1), windrows (56.1), and forest (29.0). These populations declined significantly ($F_{4,32}$ = 14.44; P < 0.01) in 2018 to within the range of total small mammal numbers recorded in the earlier period 2007–2011 (Fig. 6).

Overall mean abundance of *M. gapperi* and *M. longicaudus* indicated either significantly higher (non-overlapping 95% CIs) populations of these two voles in the piles or windrows than dispersed and/or forest sites in 2007–2011, or comparable numbers (overlapping 95% CIs) among treatment sites in 2017–2018 (Fig. 7a + b). Overall mean abundance of *N. amoenus* and total small mammals followed this same pattern as *M. longicaudus*, based on 95% confidence intervals (Fig. 8a + b).

3.2. Species richness and diversity of small mammals

Mean species richness and species diversity of the small mammal

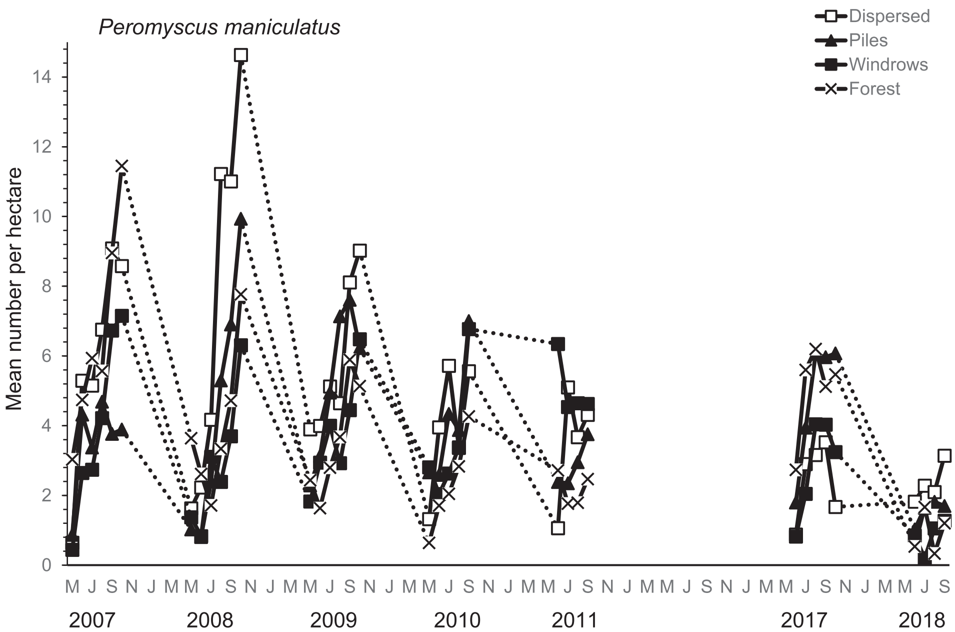

Fig. 4. Mean (n = 3 replicate sites) number of *Peromyscus maniculatus* per ha as an index based on Jolly-Seber population estimates and effective trapped area in the dispersed, piles, windrow, and forest sites during 2007–2011 and 2017–2018. Data points indicate individual trapping weeks each summer (May to October) and dots indicate winter periods when populations were not sampled.

community were significantly ($P < 0.01$) different among treatment sites in 2007–2011 (Table 1). Richness was highest (DMRT; $P = 0.05$) in the piles and windrows with the piles and dispersed also being similar, and these three treatments higher than the forest sites. A significant ($F_{4,32} = 2.67$; $P = 0.05$) time effect indicated an initial increase and then decline in richness over the 5-year period. Mean species diversity was higher (DMRT; $P = 0.05$) in the dispersed, piles, and windrow sites than forest sites and generally similar over time (Table 1). In the 2017–2018 period, both species richness and diversity were significantly ($P \le 0.05$) different among sites with the dispersed, piles, and windrow sites higher (DMRT; $P = 0.05$) than the forest sites (Table 2).

3.3. Reproduction and survival of the major species

Mean number of juvenile and total recruits of *M. gapperi* in 2007–2011 was significantly ($P \le 0.02$) different among treatment sites with higher (DMRT; $P = 0.05$) recruitment in the piles, windrow, and forest sites than in the dispersed (Table 3). These numbers declined significantly ($P \le 0.02$) with time in accordance with the generally lower populations of this microtine in 2010–2011 (Fig. 2). Mean number of successful pregnancies was also significantly ($F_{3,6} = 5.05$; $P = 0.04$) different among treatment sites with higher (DMRT; $P = 0.05$) productivity in the windrow and forest sites than in the dispersed, with similar values in the piles and dispersed (Table 3). This measure of reproductive output was consistent over time. Mean summer survival was similar ($P = 0.50$), but mean winter survival was significantly ($F_{3,6} = 6.46$; $P = 0.04$) different, among treatment sites with declining survival from the dispersed to forest sites (Table 3). On average, winter survival was 26% higher than that in summer.

In the 2017–2018 period, mean number of juvenile recruits of *M. gapperi* was similar ($P = 0.18$), but mean number of total recruits was significantly ($F_{3,6} = 6.47$; $P = 0.03$) different, among treatment sites (Table 4). There was higher (DMRT; $P = 0.05$) total recruitment in the

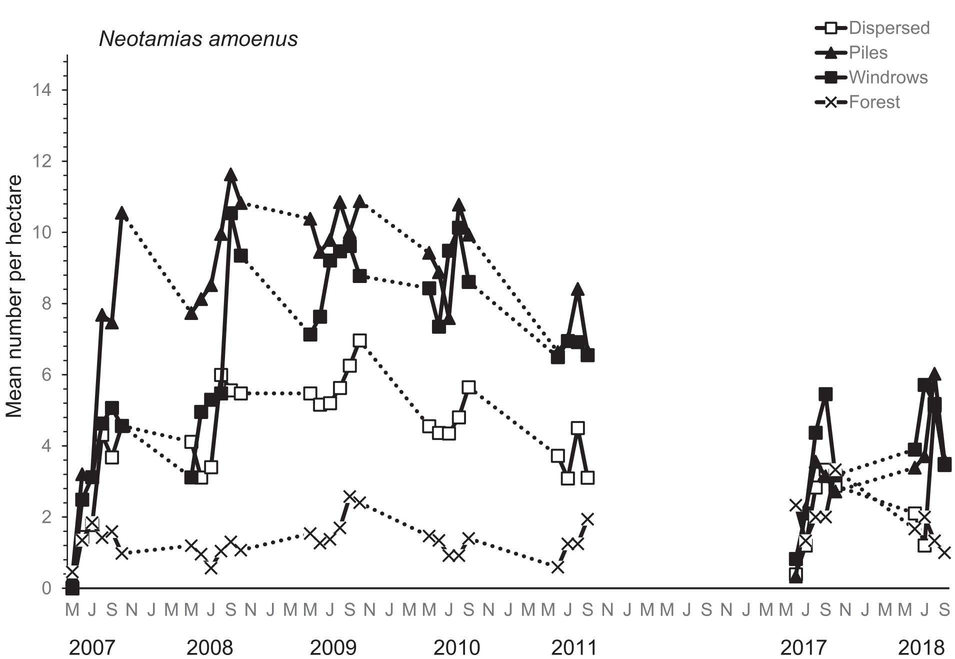

Fig. 5. Mean (n = 3 replicate sites) number of *Neotamias amoenus* per ha as an index based on Jolly-Seber population estimates and effective trapped area in the dispersed, piles, windrow, and forest sites during 2007–2011 and 2017–2018. Data points indicate individual trapping weeks each summer (May to October) and dots indicate winter periods when populations were not sampled.

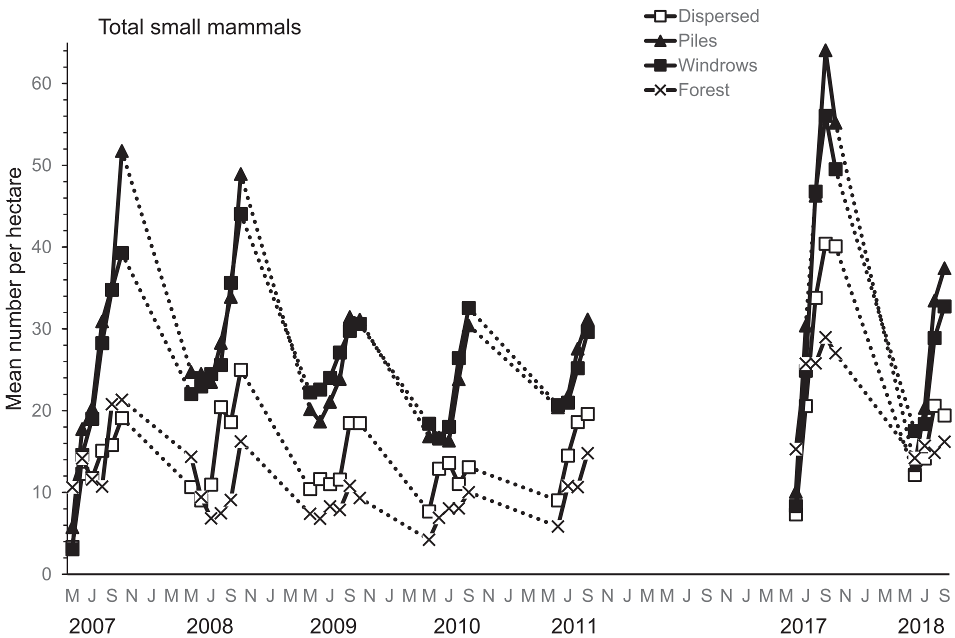

Fig. 6. Mean (n = 3 replicate sites) number of total forest-floor small mammals per ha as an index based on Jolly-Seber population estimates and effective trapped area for the major species, and number of individuals captured for the other species, in the dispersed, piles, windrow, and forest sites during 2007–2011 and 2017–2018. Data points indicate individual trapping weeks each summer (May to October) and dots indicate winter periods when populations were not sampled.

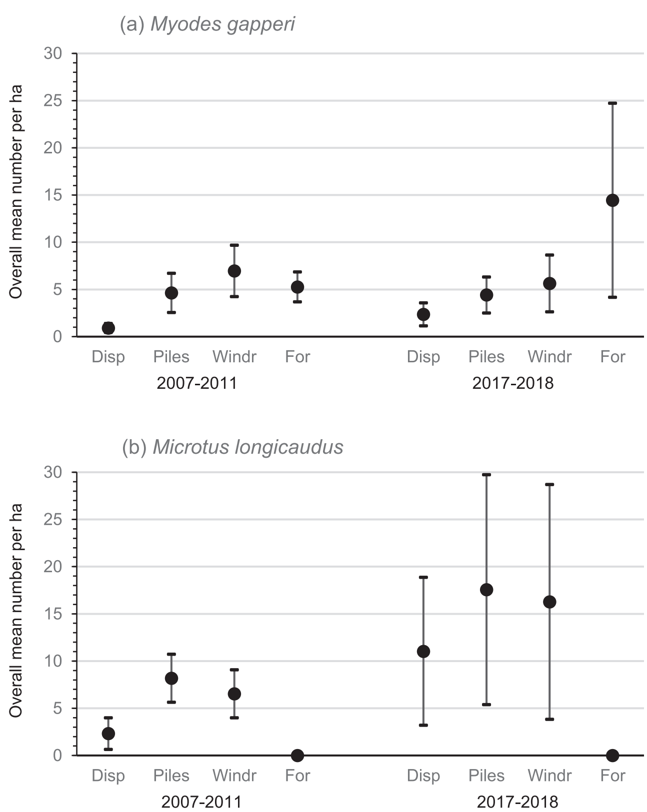

Fig. 7. Overall mean (n = 3 replicates sites × 5 years and 2 years = 15 and 6) ± 95% CIs number of (a) *M. gapperi* and (b) *M. longicaudus* in the dispersed (disp), piles, windrow (windr), and forest (for) sites during 2007–2011 and 2017–2018.

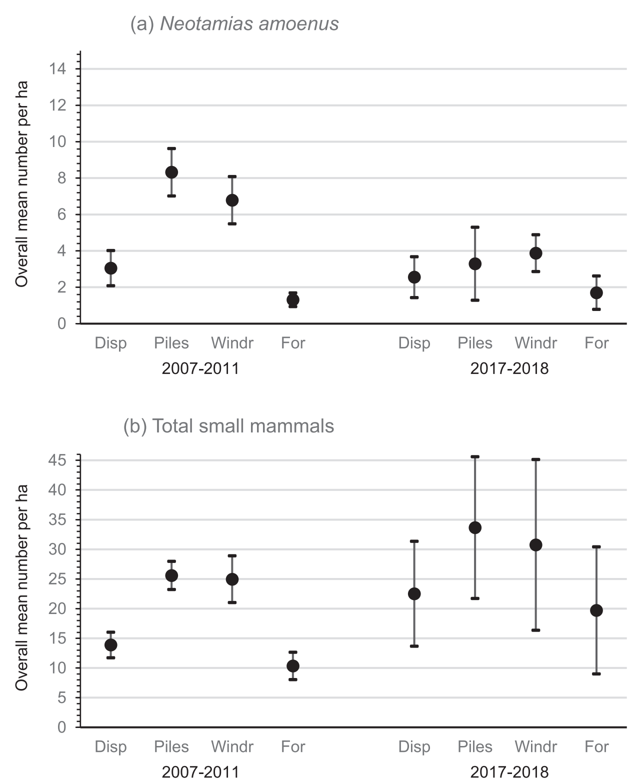

Fig. 8. Overall mean (n = 3 replicates sites × 5 years and 2 years = 15 and 6) ± 95% CIs number of (a) *N. amoenus* and (b) total small mammals in the dispersed (disp), piles, windrow (windr), and forest (for) sites during 2007–2011 and 2017–2018.

forest sites than in the dispersed and piles sites with the windrow sites being comparable to the forest and other sites. Mean number of successful pregnancies and summer and winter survival rates were similar ($P \geq 0.29$) among sites (Table 4). On average, winter survival was 28.5% higher than that in summer.

In 2007–2011, mean number of juvenile recruits of *M. longicaudus* approached significance ($P = 0.06$) and that of total recruits was significantly ($F_{2,4} = 6.75$; $P = 0.05$) different among treatment sites with higher (DMRT; $P = 0.05$) total recruitment in the piles than dispersed sites and comparable numbers in the piles and windrows (Table 3). Recruitment was generally higher (2.4 to 3.9 times, on average) for this microtine in the debris piles and windrows than dispersed sites. Mean number of successful pregnancies was similar ($P = 0.19$) among treatment sites, and along with total recruits, changed significantly ($P \leq 0.04$) with time; declining from 2007 to

Table 3
Overall mean (n = 15 or 12 replicate sites) ± SE demographic attributes for *M. gapperi*, *M. longicaudus*, and *P. maniculatus* during the 2007–2011 period among the four treatments, and results of RM-ANOVA. *F*-values identified by * were calculated using an H-F correction factor, which decreased the stated degrees of freedom due to correlation among repeated measures. Within a row, columns of mean values with different letters are significantly different by Duncan's multiple range test (DMRT), adjusted for multiple contrasts. Significant values are given in bold text.

Species + parameter	Treatment				RM-ANOVA					
					Treatment		Time		Treatment × time	
	Dispersed	Piles	Windrows	Forest						
M. gapperi					$F_{3,6}$	*P*	$F_{4,32}$	*P*	$F_{12,32}$	*P*
Juvenile recruits	1.13b ± 0.34	7.60a ± 1.93	10.67a ± 2.18	9.80a ± 1.72	8.03	**0.02**	3.44	**0.02**	0.58	0.85
Total recruits	2.93b ± 0.80	12.27a ± 2.98	17.67a ± 3.14	17.27a ± 2.48	12.97	**< 0.01**	4.12	**< 0.01**	0.78	0.67
Pregnancies	1.00b ± 0.41	3.53ab ± 0.74	5.73a ± 1.29	6.20a ± 1.10	5.05	**0.04**	0.56	0.69	0.78	0.68
					$F_{3,5}$	*P*	$F_{3,21}$	*P*	$F_{9,21}$	*P*
Summer survival	0.59 ± 0.06	0.61 ± 0.05	0.72 ± 0.03	0.58 ± 0.05	0.91	0.50	0.95	0.43	0.53	0.84
Winter survival	0.92a ± 0.02	0.90ab ± 0.01	0.87bc ± 0.02	0.84c ± 0.02	6.46	**0.04**	1.08	0.38	0.64	0.75
M. longicaudus					$F_{2,4}$	*P*	$F_{4,24}$	*P*	$F_{8,24}$	*P*
Juvenile recruits	3.87 ± 0.66	14.93 ± 2.78	13.47 ± 2.17	–	6.10	0.06	1.17*	0.34	0.31*	0.90
Total recruits	8.87b ± 1.50	24.20a ± 3.84	20.9ab ± 3.09	–	6.75	**0.05**	3.37*	**0.04**	0.55*	0.76
Pregnancies	3.13 ± 0.80	8.20 ± 2.06	6.53 ± 1.25	–	2.57	0.19	7.34	**< 0.01**	1.30	0.29
					$F_{2,4}$	*P*	$F_{3,18}$	*P*	$F_{6,18}$	*P*
Summer survival	0.73 ± 0.07	0.67 ± 0.04	0.61 ± 0.05	–	5.76	0.07	0.75	0.54	0.36	0.90
Winter survival	0.86 ± 0.03	0.76 ± 0.02	0.82 ± 0.01	–	4.99	0.08	1.25	0.32	1.83	0.15
P. maniculatus					$F_{3,6}$	*P*	$F_{4,32}$	*P*	$F_{12,32}$	*P*
Juvenile recruits	11.27 ± 2.36	15.33 ± 1.59	14.60 ± 1.25	9.87 ± 1.87	3.09	0.11	3.28	**0.02**	1.10	0.39
Total recruits	16.40 ± 2.86	18.47 ± 1.66	17.73 ± 1.42	12.87 ± 2.11	1.52	0.30	2.38	0.07	0.76	0.68
Pregnancies	2.13 ± 0.40	3.73 ± 0.80	3.27 ± 0.65	2.40 ± 0.64	0.38	0.77	1.53*	0.24	0.26*	0.97
					$F_{3,6}$	*P*	$F_{3,24}$	*P*	$F_{9,24}$	*P*
Summer survival	0.70 ± 0.05	0.59 ± 0.04	0.56 ± 0.05	0.71 ± 0.05	2.58	0.15	0.28	0.84	0.52	0.85
Winter survival	0.80 ± 0.02	0.77 ± 0.01	0.77 ± 0.02	0.80 ± 0.03	0.35	0.79	1.30	0.30	0.28	0.97

2009 and then increasing to 2011 in accordance with population changes (Table 3; Fig. 3). Mean summer (range of 61% to 73%) and winter (range of 76% to 86%) survival rates were similar among treatment sites. In 2017–2018, mean reproductive and survival attributes were similar ($P \geq 0.22$) among treatment sites for *M. longicaudus*, but there was a significant ($P = 0.02$) decline from the very high population year of 2017 to a moderate year of abundance in 2018 (Table 4; Fig. 3).

Mean reproductive and survival attributes were similar ($P \geq 0.11$) among treatment sites for *P. maniculatus* in both periods (Tables 3 and 4). There was a significant ($P \leq 0.03$) decline in mean numbers of juvenile recruits over the 2007–2011 period and for both measures of recruits in the 2017–2018 period, likely owing to declining populations (Tables 3 and 4; Fig. 4).

Table 4
Overall mean (n = 3 or 6 replicate sites) ± SE demographic attributes for *M. gapperi*, *M. longicaudus*, and *P. maniculatus* during the 2017–2018 period among the four treatments, and results of RM-ANOVA and one-way ANOVA. Within a row, columns of mean values with different letters are significantly different by Duncan's multiple range test (DMRT), adjusted for multiple contrasts. Significant values are given in bold text.

Species + parameter	Treatment				RM-ANOVA					
					Treatment		Time		Treatment × time	
	Dispersed	Piles	Windrows	Forest						
M. gapperi					$F_{3,6}$	*P*	$F_{1,8}$	*P*	$F_{3,8}$	*P*
Juvenile recruits	3.17 ± 1.01	5.33 ± 1.20	10.00 ± 2.32	13.83 ± 2.60	2.26	0.18	2.59	0.15	1.57	0.27
Total recruits	6.17b ± 1.35	10.67b ± 1.82	15.83ab ± 3.50	28.17a ± 3.17	6.47	**0.03**	0.41	0.54	0.37	0.78
Pregnancies	1.83 ± 0.79	3.33 ± 0.99	6.17 ± 1.62	15.17 ± 5.53	1.55	0.30	0.06	0.81	0.85	0.51
					$F_{3,8}$	*P*				
Summer survival	0.58 ± 0.14	0.55 ± 0.05	0.51 ± 0.03	0.65 ± 0.07	0.51	0.69				
Winter survival	0.91 ± 0.02	0.84 ± 0.03	0.88 ± 0.03	0.80 ± 0.06	1.51	0.29				
M. longicaudus					$F_{2,4}$	*P*	$F_{1,6}$	*P*	$F_{2,6}$	*P*
Juvenile recruits	18.67 ± 5.85	31.00 ± 8.41	26.17 ± 8.56	–	1.67	0.30	11.28	**0.02**	0.26	0.78
Total recruits	27.00 ± 7.55	44.67 ± 10.91	40.83 ± 12.09	–	1.78	0.28	9.45	**0.02**	0.11	0.90
Pregnancies	8.33 ± 2.23	9.67 ± 1.63	10.33 ± 4.88	–	0.05	0.96	1.81	0.23	0.24	0.80
					$F_{2,6}$	*P*				
Summer survival	0.59 ± 0.01	0.59 ± 0.08	0.71 ± 0.04	–	1.94	0.22				
Winter survival	0.75 ± 0.02	0.67 ± 0.04	0.71 ± 0.04	–	1.38	0.32				
P. maniculatus					$F_{3,6}$	*P*	$F_{1,8}$	*P*	$F_{3,8}$	*P*
Juvenile recruits	6.17 ± 1.49	6.17 ± 1.76	4.17 ± 1.68	5.00 ± 2.00	0.28	0.84	11.38	**< 0.01**	0.89	0.49
Total recruits	9.33 ± 1.82	8.83 ± 2.02	7.67 ± 1.84	9.67 ± 3.24	0.13	0.94	7.54	**0.03**	0.49	0.70
Pregnancies	1.67 ± 0.67	3.17 ± 0.87	1.67 ± 0.76	1.50 ± 0.62	0.58	0.65	0.14	0.72	0.73	0.56
					$F_{3,8}$	*P*				
Summer survival	0.54 ± 0.05	0.71 ± 0.12	0.63 ± 0.13	0.48 ± 0.15	0.74	0.56				
Winter survival	0.84 ± 0.04	0.78 ± 0.02	0.77 ± 0.02	0.76 ± 0.04	1.18	0.38				

4. Discussion

4.1. Forest-floor small mammal communities

Our results did not support the abundance part of H_1 that forest-floor small mammals would decline over time in piles and windrows of woody debris compared with sites of dispersed debris or uncut forest. Populations of *M. gapperi*, *M. longicaudus, N. amoenus*, and total small mammals were generally higher in abundance on sites with debris structures than those sites with dispersed debris in the 2007–2011 period and again in 2017–2018. Other species such as *P. maniculatus* remained similar in abundance among treatment sites. Similarly, species richness and diversity also did not fit the predictions of H_1 as these measurements were increased or maintained in the debris structures over the 12-period.

High numbers of red-backed voles in new woody debris structures was related to either the residual animals that survived clearcutting of the forest, or to high relative numbers associated with the 6-to 7-year population fluctuation of this microtine (Sullivan et al., 2017b). *M. gapperi* is a closed canopy specialist in old-growth and mature coniferous forests of western North America (Merritt, 1981) and may be considered an indicator species of "old-forest conditions" (Nordyke and Buskirk, 1991; Pearce and Venier, 2005; Boonstra and Krebs, 2012). *M. gapperi* disappears from clearcuts within a year of harvest, at least in western North America, presumably because of a loss of food, cover, and other attributes of forest stand structure (Fisher and Wilkinson, 2005; Zwolak, 2009). The increased numbers of *M. gapperi* in 2007–2009 in our debris structures was during the decline and low years of the population fluctuation for this species (Sullivan et al., 2017b). Thus, it seemed unlikely that dispersal of red-backed voles from uncut forest contributed to the relatively high numbers in the debris structures during this initial 3-year period.

Woody debris structures seem to provide habitat on new clearcuts for *M. gapperi* at comparable or higher abundance than in uncut forest. Although numbers declined after the initial three years, populations of *M. gapperi* in debris structures recovered to earlier abundance levels at 11–12 years post-construction. At least part of this recovery was likely related to the population increase and peak of *M. gapperi* in uncut forest in 2017–2018. The peak populations of 14 to 21 voles per ha in 2017–2018 in our uncut forest sites were within the range of those recorded earlier for this species over a 21-year period (Sullivan et al., 2017b).

Our study reports a multi-annual population fluctuation of *M. longicaudus* over a 12-year period, albeit with discontinuous sampling in some years. The annual peak numbers of 24 to 42 voles per ha in 2017, 11 years after harvest, were substantially higher than those recorded earlier on the same treatment sites. This result was similar to other older successional stages in Alaska (23 years) and northwest BC (9–13 years) where annual peak populations of long-tailed voles persisted at a range of densities from 30 to 70/ha (Van Horne, 1982; Sullivan et al., 1999). Several studies have suggested that long-tailed voles have an annual population cycle with peak numbers associated with relatively high abundance of grasses and forbs, at least in post-harvest sites (Sullivan and Sullivan, 2001, 2010; Craig et al., 2015). The lack of a population increase on our dispersed sites after clearcutting was surprising as *M. longicaudus* in nearby study areas were reported at relatively high numbers up to 2- to 4-years post-harvest, prior to declining to low (< 2/ha) numbers thereafter (Sullivan et al., 2008; Sullivan and Sullivan, 2010). Cattle grazing on summer range occurred on our sites in all years and may have contributed to a reduction in herbaceous food and cover for voles (Johnston and Anthony, 2008). Mean abundance of herbaceous vegetation was similar among sites in 2007–2009 (Sullivan and Sullivan, 2012), but we did not sample for abundance of grasses and herbs on our sites after this initial three years. However, cattle grazing of vegetation did not prevent the buildup in vole numbers on all three treatment sites in 2017 and maintenance of similar abundances in 2018 to those recorded in 2007–2011.

The consistently higher numbers of *M. longicaudus* in piles and windrows than in sites with dispersed woody debris indicated clearly that these structures provided sufficient habitat on new clearcuts. A positive relationship between abundance of *M. longicaudus* and volume of dispersed woody debris was also reported by Van Horne (1982) and Mason (1989). Numbers of long-tailed voles were also generally related to the number of pieces of dispersed downed wood (Sullivan and Sullivan, 2010; Craig et al., 2015). Large woody debris provides a moist microclimate for sporocarp production by hypogeous mycorrhizal fungi (Amaranthus et al., 1994), a source of conifer seeds, and invertebrates, all of which act as supplemental food sources for *M. longicaudus* (Maser et al., 2008; Smolen and Keller, 1987; Sullivan and Sullivan, 2001). Thus, it may be possible that our debris piles and windrows provided source sites for *M. longicaudus* populations to build up in numbers to high levels in 2017, with consequent dispersal of voles to other nearby habitats.

The generalist *P. maniculatus* showed no response to piles and windrows of woody debris. This result was similar to those reported for a variety of woody debris environments in other investigations (Craig et al., 2006; Hayes and Cross, 1987; Smith and Maguire, 2004; Sullivan et al., 2017a). Exceptions to this pattern were in areas of natural accumulations of debris where more *Peromyscus* occurred than at reference sites (Loeb, 1999; Steel et al., 1999). *N. amoenus* showed the first significant population response to piles and windrows in the 2007–2011 period. Other studies reported that these sciurids used woody debris for travel paths and showed little or no population response (Sullivan et al., 2017a; Waldien et al., 2006).

4.2. Reproduction and survival of the major species

Density alone of a given species may not always be an accurate indicator of habitat quality (Van Horne, 1983; Wheatley et al., 2002). To this end, reproductive and survival attributes either increased or were maintained in piles and windrow sites for *M. gapperi*, *M. longicaudus*, and *P. maniculatus*. Thus, the prediction of H_2 that reproduction and survival of these major species would decline over time in piles and windrows of woody debris, compared with sites of dispersed woody debris or uncut forest was not supported. Habitat quality in the constructed woody debris sites was presumably high enough to support these small mammal communities. These habitats were clearly not dispersal sinks (Pulliam, 1988) for subordinate animals as suggested for responses of some small mammal species, such as *Peromyscus* (Sullivan, 1979; Martell, 1983; Nelson et al., 2019) and *Myodes* (Gasperini et al., 2016; Martineau et al., 2016) to new clearcuts. In fact, our woody debris habitats may be considered sources of small mammals as a means of balanced dispersal (reviewed by Diffendorfer, 1998) to maintain these species through succession of the regenerating forest.

4.3. Study limitations

Our study design was composed of independent experimental units as per Hurlbert (1984). No voles of any species were captured on more than one grid, and hence sites were considered independent. However, this premise may not have always been accurate for wide-ranging species such as *P. maniculatus* and *N. amoenus*. Our harvested sites were the size of conventional forestry operations in the southern interior of BC and in similar regions of the Pacific Northwest. Additional spatial replicates may have strengthened the precision of our results, but was not possible in this large-scale field study. Inferences are applicable to clearcut openings of this range of sizes among forest ecosystems in southern BC. These inferences represent responses of forest-floor small mammals to habitat structures during summer and fall (May to October) only. Population changes resulting from these treatments may not have been the same during other seasons of the year. Ideally, we would have liked to have sampled populations in all years after construction of these restoration structures, but this was not feasible. Although not possible in this study, investigation of food resources within the woody debris structures would have provided additional evidence for their utility as viable habitats for small mammals.

5. Conclusions

Our study is the first to measure long-term (up to 12 years) responses of forest-floor small mammals to constructed piles and windrows of woody debris as a means of habitat retention on clearcuts. A major concern of forest and wildlife managers was that these structures would become unoccupied after the first few years post-construction as woody materials dried out and settled. This prediction was clearly not the case with respect to abundance, species richness, and diversity of these communities. In addition, demographic attributes of reproduction and survival followed the pattern of abundance for the major species: *M. gapperi*, *M. longicaudus*, and *P. maniculatus*. Piles and windrows of woody debris need to be built at an appropriate real-world scale and timing after harvest to yield these novel results in a consistent pattern.

Structures need to be at least 2–3 m in height and 5–7 m in width or diameter and ideally created at the time of forest harvesting and log processing to minimize both operational costs and excessive movement of heavy machinery over the forest floor. Depending on operational conditions and availability of woody debris, at least one windrow or a series of piles should connect patches of mature forest and riparian areas to allow small mammals and other species to access and traverse clearcut openings. This will be particularly important on conventional-sized openings (5–20 ha), but also on much larger (> 100 ha) insect-killed as well as burned forests that are salvage-harvested, at least in the Pacific Northwest of North America. To avoid the potential risk of wildfire, these structures should not be in, or near, the forest interface with human communities.

Piles and windrows of woody debris may serve, in part, as surrogates for red squirrel (*Tamiasciurus hudsonicus*) middens that contain conifer seeds, fungi, invertebrates and other resources that attract forest-floor small mammals. Small mammals may then serve as prey for marten and other mustelids who often frequent middens as foraging and resting sites. Voles, in particular, dominated the small mammal communities in these structures and are a preferred prey of many predators. This relationship provides further support for piles and windrows to act as baseline trophic structures in ecological restoration of cutover forest land.

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